Guransh Singh Kohli

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EDUCATION

University of Waterloo - Bachelor in Mathematics , Co-op Sept. 2025 - Exp. Dec'2029
Double Major: Applied Mathematics in Machine Learning & Statistics **Minor:** Computing
Coursework: Functional Programming, Data Structures & Algorithms, Object Oriented Programming, Calculus, Algebra
Certifications: AWS Cloud Practitioner, AWS AI Practitioner, Google Cloud Gen AI, Databricks AI Agent Fundamentals

EXPERIENCE

Windscribe - 8 month coop Toronto, ON
AI Engineer May 2026 - Dec. 2026

- Revamped **Garry**, an **OpenClaw**-based chatbot, across **15+ intents**, reducing human support escalations by **90%**.
- Rewrote an internal **LLM** support tool's **prompt and generation pipeline**, cutting **API token usage** by **65%**.
- Built a **Stripe dispute analyst** integrating Stripe APIs with **agentic workflows**, projected to save **$32K annually**.
- Built **LLM-powered** Stripe Radar triage that scores flagged payments against fraud signals, cutting manual work by **20%**.

Wat.ai - Offball XI(Real data & Spatial Analytics UW Design Team) Waterloo, ON
Data Science Engineer Sept. 2026 - Present

- Building a Python pipeline to ingest and process **100k+ tracking records** from 2022 FIFA World Cup data.
- Standardizing data across **64 matches and 800+ player-tracking fields** to align World Cup data with GRF.
- Engineering **7+ spatial features** using Pandas and NumPy to prepare tracking data for machine learning experiments.
- Automating data validation and transformation across **8 processing steps**, producing ML-ready datasets for counterfactual run-value modelling.

Savi Finance Toronto, ON
Software Developer Nov. 2025 - May 2026

- Built **RESTful APIs in Go** with MySQL powering trip-logging and expense-tracking workflows for **1.5K+ users**.
- Redesigned **relational schemas** & optimized queries, reducing core expense-tracking query from **2s to 400ms (5x faster)**.
- Resolved recurring **data-integrity issues** through improved request validation across production services.

TECHNICAL PROJECTS

Wanderers.space Prototype Waterloo, ON
Founding Engineer Jan. 2026 - Present

- Built **Wanderers**, a Next.js campus social platform serving **46 MAU** of verified Waterloo students.
- Architected **REST APIs, Realtime pub/sub, and live chat** for activity matching, feeds, & real-time group interactions.
- Integrated **Gemini AI, Google Maps, and K-means clustering** for activity creation, & personalized recommendations.
- Pitched at Velocity receiving mentorship from **2 Velocity UW incubator mentors** on product development.

QuantPilot - Stocks Intelligence Platform *LangGraph, Kafka, PySpark, FastAPI, PostgreSQL* GitHub

- Architected an **event-driven research platform** for market events, technology trends, and second-order impact analysis.
- Engineered **Kafka + PySpark pipelines** for scalable ingestion and processing of financial, news, and research data.
- Built a **Neo4j knowledge graph** mapping companies, technologies, investments, and partnerships for market analysis.
- Developed multi-agent research workflows with source verification and PostgreSQL for traceable, evidence-backed analysis.

RoadSafe AI - US Road Safety Analytics *Python, PostgreSQL, Docker, Pandas, NumPy,Scikit-learn, Tableau* GitHub

- Modeled **100K US crash records** into a dbt + Postgres **star schema** (fact + 4 dims) with no imputation in raw load.
- Showed national **rain-vs-clear severity gap** is state composition, not weather (California **+0.46 pp**; Florida **+9.45 pp**).
- Isolated **junction proximity** at **∼27.9%** severe across three baselines, the one condition surviving base-rate correction.
- Set **80.55% majority-class floor** that weather/time features failed to beat; published findings to a **Tableau Dashboard**.

SKILLS

Languages: Python, Go, C++, C, JavaScript, TypeScript, SQL, Racket
Backend & Web: FastAPI, REST APIs, GraphQL,Next.js, React, PostgreSQL, Redis, Supabase, Docker, System Design
AI & ML: PyTorch, Scikit-learn, NumPy, Pandas, LangGraph, MCP, RAG, XGBoost, Statistical Analysis
Tools & Others: Git, GitHub, OpenClaw, CrewAI, Redis, Linux